

5.5.1 Integration by Substitution (Indefinite Integrals)

Motivation: Suppose we wanted to determine $\int 2x\sqrt{1+x^2} dx$.

It may at first be helpful to note that if we were to differentiate $(1+x^2)^p$ a $2x$ would appear from the Chain Rule. With this in mind we may be able to reverse the order of differentiate successfully.

We suspect that $\int 2x\sqrt{1+x^2} dx = \frac{2}{3}(1+x^2)^{3/2} + C$.

We can check our answer by differentiation.

$$\frac{d}{dx} \left[\frac{2}{3}(1+x^2)^{3/2} + C \right] = \frac{3}{2} \cdot \frac{2}{3}(1+x^2)^{1/2} \cdot 2x = 2x\sqrt{1+x^2} \quad \text{It works!!!}$$

We have essentially “reversed” the Chain Rule

The process of “reversing” the Chain Rule can be a bit difficult at times. We now introduce some notation to make this process a bit easier.

Let $u = 1+x^2$, then $\frac{du}{dx} = 2x \Rightarrow du = 2x dx$. We substitute these values into our integral and obtain:

$$\int 2x\sqrt{1+x^2} dx = \int \sqrt{u} du = \int u^{1/2} du = \frac{2}{3} u^{3/2} + C = \frac{2}{3}(1+x^2)^{3/2} + C$$

Terminology: We call this approach of “reversing” the Chain Rule a ***u*-substitution**.

Theorem: Suppose g is a differentiable function whose range is an I , f is continuous on I , and $u = g(x) \Rightarrow du = g'(x) dx$, then

$$\int f(g(x)) \cdot g'(x) dx = \int f(u) du$$

Note: In the motivation above we had $f(x) = \sqrt{x}$ and $g(x) = 1+x^2$.

Note: The most difficult part of the u -substitution process is determining what expression to let u equal. Unfortunately, there is no general algorithm for determining the u , but one can get a feel for what to let u equal by experiencing many problems.

Example 1: Find $\int x^3 \cos(x^4 + 2) dx$

Let $u = x^4 + 2$, then $du = 4x^3 dx \Rightarrow \frac{1}{4} du = x^3 dx$.

Therefore $\int x^3 \cos(x^4 + 2) dx = \int \cos(u) \cdot \frac{1}{4} du = \frac{1}{4} \int \cos(u) du = \frac{1}{4} \sin(u) + C = \frac{1}{4} \sin(x^4 + 2) + C$.

Example 2: Find $\int \frac{x}{\sqrt{1-4x^2}} dx$

Let $u = 1 - 4x^2$, then $du = -8x dx \Rightarrow -\frac{1}{8} du = x dx$.

Therefore $\int \frac{x}{\sqrt{1-4x^2}} dx = \int \frac{1}{\sqrt{u}} \cdot \frac{-1}{8} du = -\frac{1}{8} \int \frac{1}{\sqrt{u}} du = -\frac{1}{8} \int u^{-1/2} du = -\frac{1}{8} \cdot 2u^{1/2} + C = -\frac{1}{4} \sqrt{1-4x^2} + C$.

Example 3: Find $\int \tan(x) dx$

It is not clear how a u -substitution could be used here, so we will attempt to rewrite the integral. We note that:

$$\int \tan(x) dx = \int \frac{\sin(x)}{\cos(x)} dx$$

Let $u = \cos(x)$, then $du = -\sin(x) dx \Rightarrow -du = \sin(x) dx$.

Therefore $\int \frac{\sin(x)}{\cos(x)} dx = \int -\frac{1}{u} du = -\ln|u| + C = -\ln|\cos(x)| + C = \ln|\sec(x)| + C$.

Note: The result that $\int \tan(x) dx = -\ln|\cos(x)| + C = \ln|\sec(x)| + C$ is very important in calculus II.

Note: In the above example we may have been tempted to try the u substitution: $u = \sin(x) \Rightarrow du = \cos(x) dx$. However, we notice that there is no nice way to manipulate the right side of the later equation to get an expression in the integral (since the integral contains $\frac{1}{\cos(x)} dx$ not $\cos(x) dx$). This illustrates that some u -substitutions will be unsuccessful, and we have to be willing to try different u -substitutions if our first attempt does not work.

Example 4: Find $\int x^5 \sqrt{1+x^2} dx$.

Let $u = 1+x^2$, then $du = 2x dx \Rightarrow \frac{1}{2} du = x dx$.

We also note that if $u = 1+x^2$ then $x^2 = u-1 \Rightarrow x^4 = (u-1)^2$.

Therefore,

$$\begin{aligned} \int x^5 \sqrt{1+x^2} dx &= \int (u-1)^2 \sqrt{u} \cdot \frac{1}{2} du = \frac{1}{2} \int (u^2 - 2u + 1) u^{1/2} du = \frac{1}{2} \int (u^{5/2} - 2u^{3/2} + u^{1/2}) du \\ &= \frac{1}{2} \left[\frac{2}{7} u^{7/2} - 2 \cdot \frac{2}{5} u^{5/2} + \frac{2}{3} u^{3/2} + C \right] \\ &= \frac{1}{7} u^{7/2} - \frac{2}{5} u^{5/2} + \frac{1}{3} u^{3/2} + C \\ &= \frac{1}{7} (1+x^2)^{7/2} - \frac{2}{5} (1+x^2)^{5/2} + \frac{1}{3} (1+x^2)^{3/2} + C \end{aligned}$$

Example 5: Determine each of the following.

a) Find $\int \frac{\sin(\sqrt{x})}{\sqrt{x}} dx$.

Let $u = \sqrt{x}$, then $du = \frac{1}{2\sqrt{x}} dx \Rightarrow 2du = \frac{1}{\sqrt{x}} dx$.

$$\text{Therefore } \int \frac{\sin(\sqrt{x})}{\sqrt{x}} dx = \int \frac{\sin(u)}{1} \cdot 2du = 2 \int \sin(u) du = -2 \cos(u) + C = -2 \cos(\sqrt{x}) + C.$$

b) Find $\int e^{-5r} dr$.

Let $u = -5r$, then $du = -5dr \Rightarrow -\frac{1}{5} du = dr$.

$$\text{Therefore } \int e^{-5r} dr = \int e^u \cdot \frac{-1}{5} du = -\frac{1}{5} \int e^u du = -\frac{1}{5} e^u + C = -\frac{1}{5} e^{-5r} + C.$$

c) Find $\int \cot(x) dx$.

It is not clear how a u -substitution could be used here, so we will attempt to rewrite the integral. We note that:

$$\int \cot(x) dx = \int \frac{\cos(x)}{\sin(x)} dx$$

Let $u = \sin(x)$, then $du = \cos(x) dx$.

$$\text{Therefore } \int \frac{\cos(x)}{\sin(x)} dx = \int \frac{1}{u} du = \ln|u| + C = \ln|\sin(x)| + C.$$

d) Find $\int \frac{2^t}{2^t+3} dt$.

Let $u = 2^t + 3$, then $du = 2^t \cdot \ln(2) dt \Rightarrow \frac{1}{\ln(2)} du = 2^t dt$.

Therefore $\int \frac{2^t}{2^t+3} dt = \int \frac{1}{u} \cdot \frac{1}{\ln(2)} dt = \frac{1}{\ln(2)} \int \frac{1}{u} du = \frac{1}{\ln(2)} \cdot \ln|u| + C = \frac{1}{\ln(2)} \cdot \ln(2^t+3) + C$.

Note: we don't need absolute value bars in our final answer since 2^t+3 is always positive.

e) Find $\int \frac{x}{1+x^4} dx$.

Let $u = x^2$, then $du = 2x dx \Rightarrow \frac{1}{2} du = x dx$.

Therefore $\int \frac{x}{1+x^4} dx = \int \frac{1}{1+u^2} \cdot \frac{1}{2} du = \frac{1}{2} \int \frac{1}{1+u^2} du = \frac{1}{2} \tan^{-1}(u) + C = \frac{1}{2} \tan^{-1}(x^2) + C$.

f) Find $\int x^{13} (x^7+9)^{19} dx$

Let $u = x^7 + 9$, then $du = 7x^6 dx \Rightarrow \frac{1}{7} du = x^6 dx$.

We also note that if $u = x^7 + 9$ then $x^7 = u - 9$.

Therefore,

$$\begin{aligned} \int x^{13} (x^7+9)^{19} dx &= \int (u-9)u^{19} \cdot \frac{1}{7} du = \frac{1}{7} \int (u^{20} - 9u^{19}) du \\ &= \frac{1}{7} \left[\frac{1}{21} u^{21} - 9 \cdot \frac{1}{20} u^{20} + C \right] \\ &= \frac{1}{147} u^{21} - \frac{9}{140} u^{20} + C \\ &= \frac{1}{147} (x^7+9)^{21} - \frac{9}{140} (x^7+9)^{20} + C \end{aligned}$$